PROBLEM SET 6 DUE 5.5

1. Let $\left\{P_i\right\}$, $\left\{Q_i\right\}$ be families of A-modules. Choose *one* of the following statements to prove:

(1) $\bigoplus_{i\in I} P_i$ is projective if and only if each P_i is projective.
(2) $\prod_{i\in I} Q_i$ is injective if and only if each Q_i is injective.

2. Let A be a ring, $\mathfrak{a}$ an ideal of A. Show that $A/\mathfrak{a}$ is a projective A-module if and only if $\mathfrak{a} = Ae$ for some idempotent e.

3. Let P be an A-module. Define $P^* \coloneqq \mathrm{Hom}_A(P, A)$. Prove the following statements:

(1) P is projective if and only if there exist a family of elements $\left\{x_i : i \in I\right\}$ in P and a family of elements $\left\{f_i : i \in I\right\}$ in P^* (indexed by the same set I) such that for any $x \in P$, one has $f_i(x) = 0$ for almost all i, and
$$x = \sum_{i\in I} f_i(x)x_i.$$
(2) P is finitely generated and projective if and only if there exists an integer $n \geqslant 1$, elements $x_1, \ldots, x_n \in P$ and $f_1, \ldots, f_n \in P^*$ such that for every $x \in P$, one has
$$x = \sum_{i=1}^{n} f_i(x)x_i.$$
We call $\left\{x_i\right\}_{i=1}^n$ and $\left\{f_i\right\}_{i=1}^n$ a **dual basis** for P.
(3) If P is finitely generated and projective, then P^* is finitely generated and projective.
(4) If P is finitely generated and projective, then $P^{**} \cong P$ as A-modules.

4. Let A be an integral domain with field of fractions K, and let $\mathfrak{a}$ be a nonzero ideal of A. Define
$$\mathfrak{a}^{-1} \coloneqq \{x \in K : x\mathfrak{a} \subseteq A\}.$$
Prove the following statements:

(1) The map
$$\mathfrak{a}^{-1} \longrightarrow \mathrm{Hom}_A(\mathfrak{a}, A); \quad x \longmapsto (a \mapsto ax)$$
is an isomorphism of A-modules.
(2) $\mathfrak{a}$ is a projective A-module if and only if
$$\mathfrak{a}\mathfrak{a}^{-1} = A$$
where $\mathfrak{a}\mathfrak{a}^{-1}$ is the A-submodule of K generated by xy for $x \in \mathfrak{a}$ and $y \in \mathfrak{a}^{-1}$. Moreover, if $\mathfrak{a}$ is projective, it is finitely generated.
(3) Let $A = \mathbb{Z}[\sqrt{-5}]$ and $\mathfrak{p} = (2, 1 + \sqrt{-5}) \subseteq A$. Show that $\mathfrak{p}$ is a non-free projective A-module.

5. Let A be an integral domain with field of fractions K.

(1) Show that K is an injective A-module.
(2) Let $A = \mathbb{Z}[x]$ and $K = \mathbb{Q}(x)$. Show that $\mathbb{Q}(x)/\mathbb{Z}[x]$ is *not* an injective $\mathbb{Z}[x]$-module.

This exercise shows that divisible modules need not be injective and quotient of an injective module need not be injective.

6. Let $n \geqslant 2$ be an integer and $d|n$. Determine (with proof) which divisors d have the property that $\mathbb{Z}/d\mathbb{Z}$ is an injective $\mathbb{Z}/n\mathbb{Z}$-module.

7. This is a continuation of Problem Set **5**, Problem **10**.
Let $(F, G, \alpha) : A\textbf{-Mod} \to B\textbf{-Mod}$ be an adjunction of additive functors.
Choose *one* of the following statements to prove:

(1) If F transforms injections of A-modules into injections of B-modules (i.e. F is exact), then G transforms injective B-modules into injective A-modules.

(2) If G transforms surjections of B-modules into surjections of A-modules (i.e. G is exact), then F transforms projective A-modules into projective B-modules.

8. Let A be a ring, M an A-module. Define

$$M^\vee \coloneqq \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}).$$

Show that M is a flat A-module if and only if $M^\vee$ is an injective A-module.

9. Let $\{M_i\}_{i\in I}$ be a family of A-modules. Show that $\bigoplus_{i\in I} M_i$ is flat if and only if each M_i is flat.

10. Prove the following statements:

(1) If B is a flat A-algebra and N is a flat B-module, then N is flat as an A-module.
(2) If B is a flat A-algebra and C is a flat B-algebra, then C is a flat A-algebra.

11. Let M be a flat A-module and $\mathfrak{a}, \mathfrak{b}$ be ideals of A. Show that $\mathfrak{a}M \cap \mathfrak{b}M = (\mathfrak{a} \cap \mathfrak{b})M$.

Hint: $0 \to \mathfrak{a} \cap \mathfrak{b} \to \mathfrak{a} \oplus \mathfrak{b} \to \mathfrak{a} + \mathfrak{b} \to 0$.

12. Let $A = k[x, y]$ where k is a field. This exercise sketches two more proofs of the fact that $\mathfrak{m} = (x, y)$ is **not** a flat A-module.

(1) If $\mathfrak{m}$ is a flat A-module, then the map

$$f : \mathfrak{m} \otimes_A \mathfrak{m} \longrightarrow \mathfrak{m}; \quad \sum_i a_i \otimes b_i \longmapsto \sum_i a_i b_i$$

is injective. Clearly, $f(x \otimes y - y \otimes x) = 0$. Show that $x \otimes y - y \otimes x \neq 0$ in $\mathfrak{m} \otimes_A \mathfrak{m}$.

Hint: It suffices to show $\bar{x} \otimes \bar{y} - \bar{y} \otimes \bar{x} \neq 0$ in $\mathfrak{m}/\mathfrak{m}^2 \otimes_A \mathfrak{m}/\mathfrak{m}^2$. Since $\mathfrak{m}/\mathfrak{m}^2$ is naturally an $A/\mathfrak{m}$-module, we have $\mathfrak{m}/\mathfrak{m}^2 \otimes_A \mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{m}/\mathfrak{m}^2 \otimes_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2$ where $A/\mathfrak{m} \cong k$ and $\mathfrak{m}/\mathfrak{m}^2$ is a 2-dim'l k-vector space with basis $\bar{x}$, $\bar{y}$.

(2) Put $\mathfrak{a} = (x)$ and $\mathfrak{b} = (y)$. Show that $\mathfrak{a}\mathfrak{m} \cap \mathfrak{b}\mathfrak{m} \neq (\mathfrak{a} \cap \mathfrak{b})\mathfrak{m}$.

13. Let $A = k[x, y]/(y^2 - x^3)$ and $B = k[t]$ where k is a field, and let $f : A \to B$ be the k-algebra homomorphism such that $f(\bar{x}) = t^2$ and $f(\bar{y}) = t^3$. Prove the following statements:

(1) f is injective.
(2) f is not flat, i.e. B is not a flat A-algebra.